

3D Spatiotemporal Tensor (DynamicRaster Assembly)

Time Slice t=0
(Start Timestamp)
Interpolated Rain Grid

Spatial Mesh (2D)

X-Axis: Easting (Width)
Y-Axis: Northing (Height)
Z-Axis: Value (mm/h)

- Resolution optimized via `_downsample_dem`

+ Δt

Temporal Filtering
Handles missing active
measurement windows

Time Slice t=1
[Data Integrity: Fallback to zeros]
No active stations

+ Δt

Time Slice t=2
Interpolated Rain Grid

+ Δt

...
Interpolated Rain Grid

+ Δt

Time Slice t=N
(Simulation End)
Interpolated Rain Grid

Δt = Time Step